COMPUTER ENGINEERING

QUEZON CITY

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| **COURSE CODE/TITLE:** | CPE 010 - Data Structures and Algorithms | **SECTION:** | CPE21S2 |
| **DATE PERFORMED:** | 08/14/26 | **DATE SUBMITTED:** | 08/20/26 |
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## ACTIVITY NO. 6.1 Linear and Binary Search

### 1. PROCEDURE OUTPUT

What procedures did you perform?

1. **What is a search tree in data structures?**
   - Search trees form the backbone of efficient data management by structuring information hierarchically. AVL trees, Red-black trees, and Binary Search Trees each offer unique tradeoffs between implementation complexity and performance guarantees, making them invaluable for different scenartios.
2. **What are the Different types of search algorithm in data structures? Differentiate each type of search.**
   - Three primary search strategies exist in computer science. Linear search methodically checks every element sequentially, resulting in O(n) complexity but requiring no preprocessing. Binary search leverages sorted data by repeatedly halving the search space, delivering superior O(log n) performance. Hash search bypasses sequential checking entirely, using hash functions to achieve O(1) theoretical access time, though collision handling adds complexity.
3. **What operations / implementations can be performed using binary and linear search operations?**
   - Search mechanisms power essential database operations including element retrieval, data
   - insertion, and record deletion. Beyond these basics, they facilitate range searches across data intervals, identify extreme values like maximums and minimums, and support sorting routines. Tree structures also allow traversal operations that expose data in sorter order naturally.
4. **What are the advantages in using binary search tree as data structure?**
   - Binary Search Trees provides fast O(log n) searching, inserting, and deleting while keeping data sorted. They handle changes to data easily, unlike sorted arrays, making them practical for applications that constantly add or remove information.
5. **Give an example program using binary search and Linear search.**

villaruellinearandbinary.cpp

C: > Users > Administrator > Desktop > CPE21S1 - Villaruel > villaruellinearandbinary.cpp > main()

```cpp
#include <iostream>
//linear search
int linear_search (int arr[] , int n, int target){
    for(int i = 0; i < n; i++)
        if (arr[i] == target) return i; //return index
    return -1; //if not found
}

//binary search
int binary_search(int arr[], int n, int target){
    int left = 0, right = n - 1;
    while (left <= right){
        int mid = (left + right) / 2;
        if (arr[mid] == target) return mid; //found target
        else if (arr[mid] < target) left = mid + 1; //search right half
        else right = mid - 1; //search left half
    }
    return -1;
}

int main(){
    int arr[] = {10, 20, 30, 40, 50};
    //testing
    std::cout<<linear_search(arr, 5, 30)<<std::endl;
    std::cout<<binary_search(arr, 5, 30)<<std::endl;
}
```

```
PS C:\Users\Administrator\Desktop\CPE21S1 - Villaruel>
2
2
```

**Observation:** I tried creating linear and binary search implementations from what I saw while researching it. I used that as my basis to understand how each algorithm works. Linear search by checking every element, and binary search by repeatedly dividing the sorted array in half. Both return index 2 for the target value 30

## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:** _____

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:** _____

COMPUTER ENGINEERING

QUEZON CITY

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:**

*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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